

◆ HOTS Question 1

Integrated Reactive and Deliberative Control Architecture for Autonomous Mobile Robot

A mobile robot is required to operate in a semi-structured indoor industrial environment where it must:

- Follow predefined floor markings
- Avoid static and dynamic obstacles
- Recover from temporary loss of path
- Operate under varying lighting and sensor noise

Question:

Design and critically evaluate a **ROS-based hybrid control architecture** that integrates:

- Line-following control
- Obstacle avoidance
- Behavior arbitration
- Fault recovery mechanisms

Your response must address the following:

1. Architecture Design

- Propose a modular ROS node architecture.
- Define communication mechanisms (topics/services/actions).
- Justify modular vs monolithic design in terms of scalability, maintainability, and fault isolation.

2. Control Strategy Evaluation

- Compare PID-based control, heuristic control, and model-based control for line following.
- Analyze stability, responsiveness, and robustness under noise.
- Identify failure cases and performance trade-offs.

3. Behavior Arbitration

- Compare priority-based, behavior-based (subsumption), and state-machine-based arbitration.
- Justify your chosen arbitration logic using safety and performance reasoning.
- Provide a formal state-transition model.

4. System Robustness Analysis

- Identify potential failure modes (sensor drift, actuator delay, communication latency).
- Propose mitigation strategies supported by control theory or probabilistic reasoning.
- Discuss real-time execution constraints.

Expected Cognitive Level:

Analyze, Evaluate, Synthesize, Justify

◆ HOTS Question 2

Comparative Design and Deployment Strategy for SLAM-Based Autonomous Navigation

An autonomous service robot must navigate a multi-floor academic building characterized by:

- Long repetitive corridors
- Glass and reflective surfaces
- Dynamic human traffic
- Intermittent Wi-Fi connectivity
- Limited onboard computation

Question:

Develop a **complete mapping, localization, and navigation strategy**, and critically evaluate competing approaches before selecting a final system configuration.

Your response must include:

1. SLAM Algorithm Comparison

- Compare at least three SLAM approaches (e.g., LiDAR-based, visual SLAM, graph-based methods).
- Analyze drift behavior, loop closure performance, computational complexity, and robustness in feature-poor environments.
- Justify algorithm selection for both LiDAR-only and vision-based configurations.

2. Localization Under Dynamic Conditions

- Compare Particle Filter-based localization and Kalman Filter-based localization.
- Analyze convergence behavior, resilience to incorrect map updates, and computational scalability.
- Propose a divergence detection and recovery strategy.

3. Navigation Stack Design

- Design a two-layer planner (global + local).
- Justify planner selection (graph-based vs sampling-based).
- Analyze costmap configuration, obstacle inflation radius, and replanning strategies.

4. Deployment-Level Trade-Off Analysis

- Evaluate effects of CPU constraints, battery limits, and communication latency.
- Propose performance metrics (e.g., localization RMSE, path efficiency, collision margin).
- Discuss scalability to multi-robot coordination and map consistency.

Expected Cognitive Level:

Evaluate, Compare, Design, Optimize, Defend